

HAT-P-18b

R-2369

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_220624 · created 2026-08-20T00:18:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459754.87021 +/- 0.00045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.046
Transit depth (Rp/Rs)^2	2.59 +/- 1.47 [%]
Orbital Inclination (inc)	89.95 +/- 1.65
Airmass coefficient 1 (a1)	1.027 +/- 0.019
Airmass coefficient 2 (a2)	-0.014 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.84 %
Best Comparison Star	#3 - [313, 430]
Optimal Method	PSF photometry
Transit Duration (day)	0.0716 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.161 ± 0.046 vs 0.1356 (deviation 0.54σ)
Duration fit vs published	0.0716 d vs 0.1129 d (deviation 4.02σ)
Mid-time O–C	0.8 min
Depth SNR	3.4
Residual scatter	1.84 %

Light curve

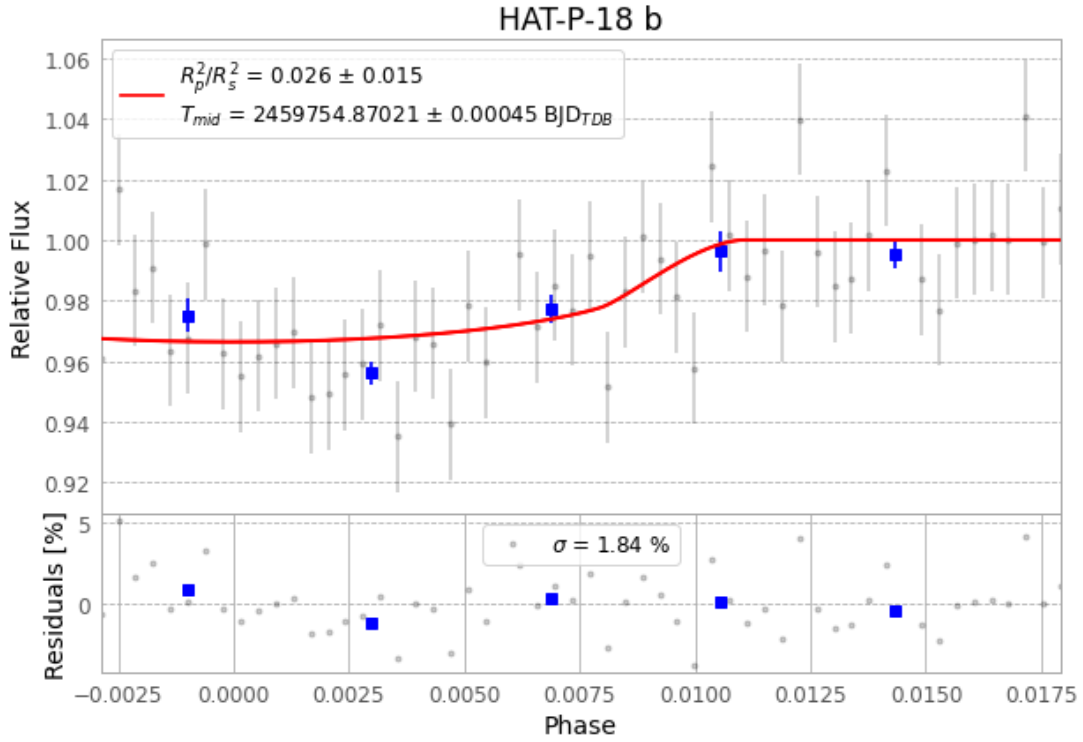


Figure 1 — FinalLightCurve_HAT-P-18 b_2022-06-24.png (SHA-256 8731a13461b544ed...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 297], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 58af66fe6141b882...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), HATP-18220624082415.FITS → HATP-18220624110914.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-220624.log (28a68ec27f04...), FinalLightCurve_HAT-P-18 b_2022-06-24.png (8731a13461b5...), FinalLightCurve_HAT-P-18 b_2022-06-24.csv (a12251b0e667...)

Compute resources

Wall time 522.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:18Z.